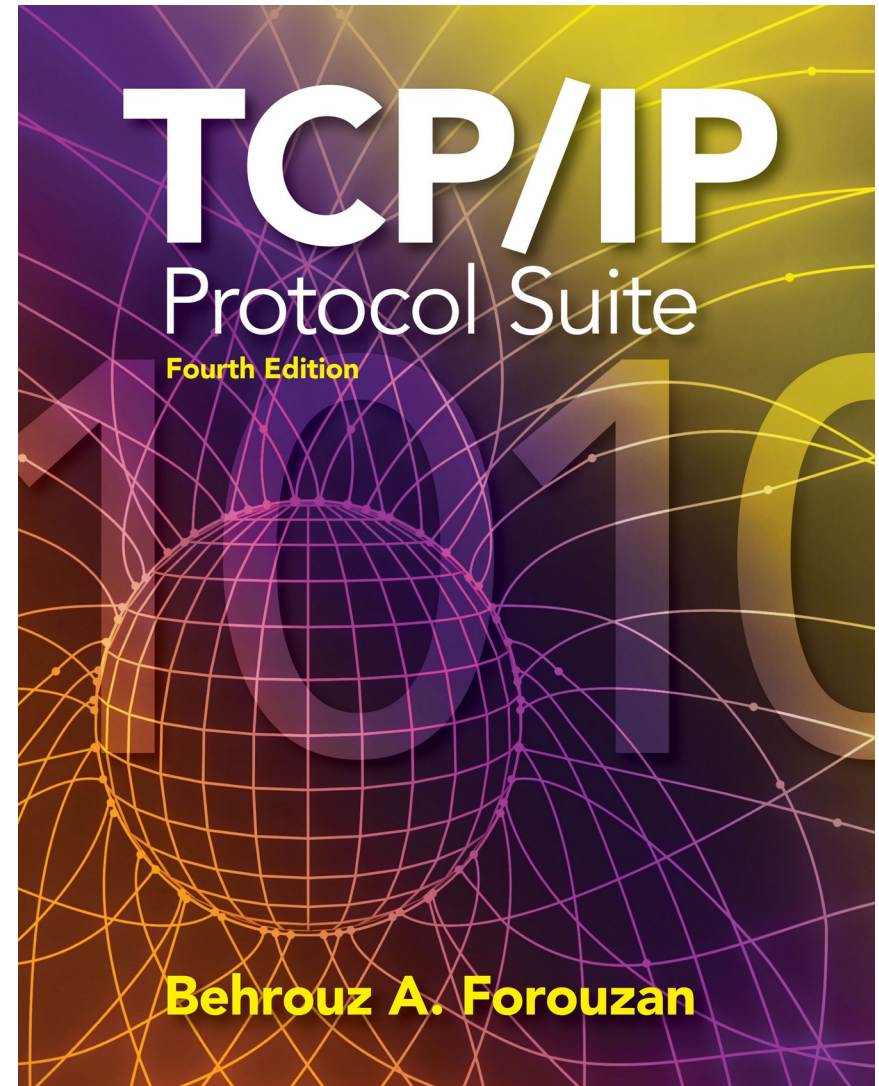




OBJECTIVES:

To give the format of an IPv6 datagram composed of a base header and a payload.

To discuss different fields used in an IPv6 datagram based header and compare them with the fields in IPv4 datagram.

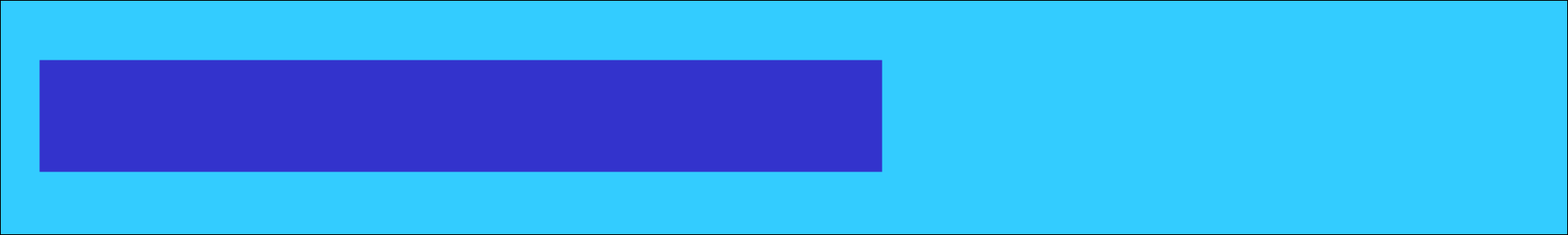
To show how the options in IPv4 header are implemented using the extension header in IPv6.

To show how security is implemented in IPv6.

To discuss three strategies used to handle the transition from IPv4 to IPv6: dual stack, tunneling, and header translation.

Chapter Outline





In this introductory section, we discuss two topics: rationale for a new protocol and the reasons for delayed adoption.

Rationale for Change

Reason for Delay in Adoption



The IPv6 packet is shown in Figure 27.1. Each packet is composed of a mandatory base header followed by the payload. The payload consists of two parts: optional extension headers and data from an upper layer. The base header occupies 40 bytes, whereas the extension headers and data from the upper layer contain up to 65,535 bytes of information.

Base Header

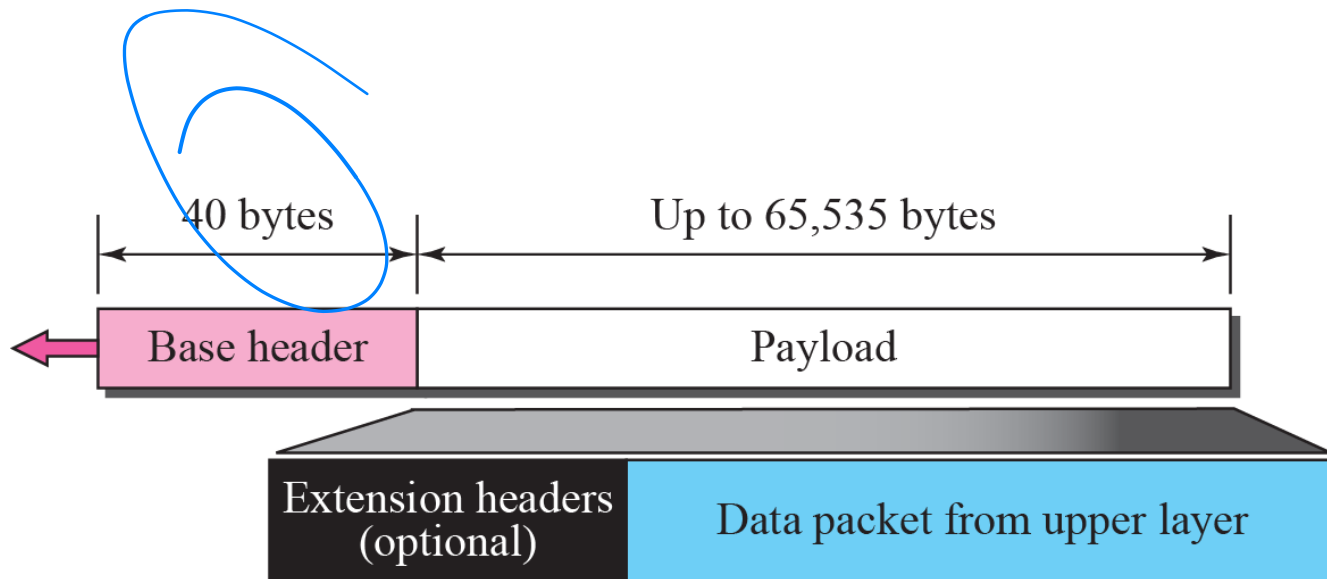
Flow Label

Comparison between IPv4 and IPv6 Headers

Extension Headers

Comparison between IPv4 and IPv6 Options

Figure 27.1 *IPv6 datagram*



📦 Fields of the IPv6 Header

1 Version (4 bits)

- Value = 6
- Identifies the protocol as IPv6
- 📌 Same purpose as IPv4 version field

2 Traffic Class (8 bits)

- Used for Quality of Service (QoS)
- Helps routers prioritize packets (voice, video, etc.)
- 📌 Similar to IPv4 Type of Service (ToS)

3 Flow Label (20 bits)

- Identifies packets belonging to the same flow
- Enables special handling for real-time traffic (VoIP, video)
- 📌 New in IPv6 (not present in IPv4)

4 Payload Length (16 bits)

- Length of payload + extension headers
- Excludes the IPv6 base header
- 📌 IPv4 length included header + data

5 Next Header (8 bits)

- Indicates what comes next
 - Another IPv6 extension header
 - Or a transport protocol (TCP, UDP, ICMPv6)

📌 Replaces:

- IPv4 Protocol field
- IPv4 Options

Examples:

Value	Meaning
6	TCP
17	UDP
58	ICMPv6

6 Hop Limit (8 bits)

- Maximum number of hops allowed
- Decremented by each router
- Packet discarded when it reaches 0
- 📌 Same as IPv4 TTL

7 Source Address (128 bits)

- IPv6 address of the sender
- 📌 Much larger than IPv4's 32-bit address

8 Destination Address (128 bits)

- IPv6 address of the receiver

📦 Why IPv6 Header Is Simpler than IPv4

❌ Removed from IPv6:

- Header checksum
- Fragmentation fields
- Options field

📌 These are handled using extension headers

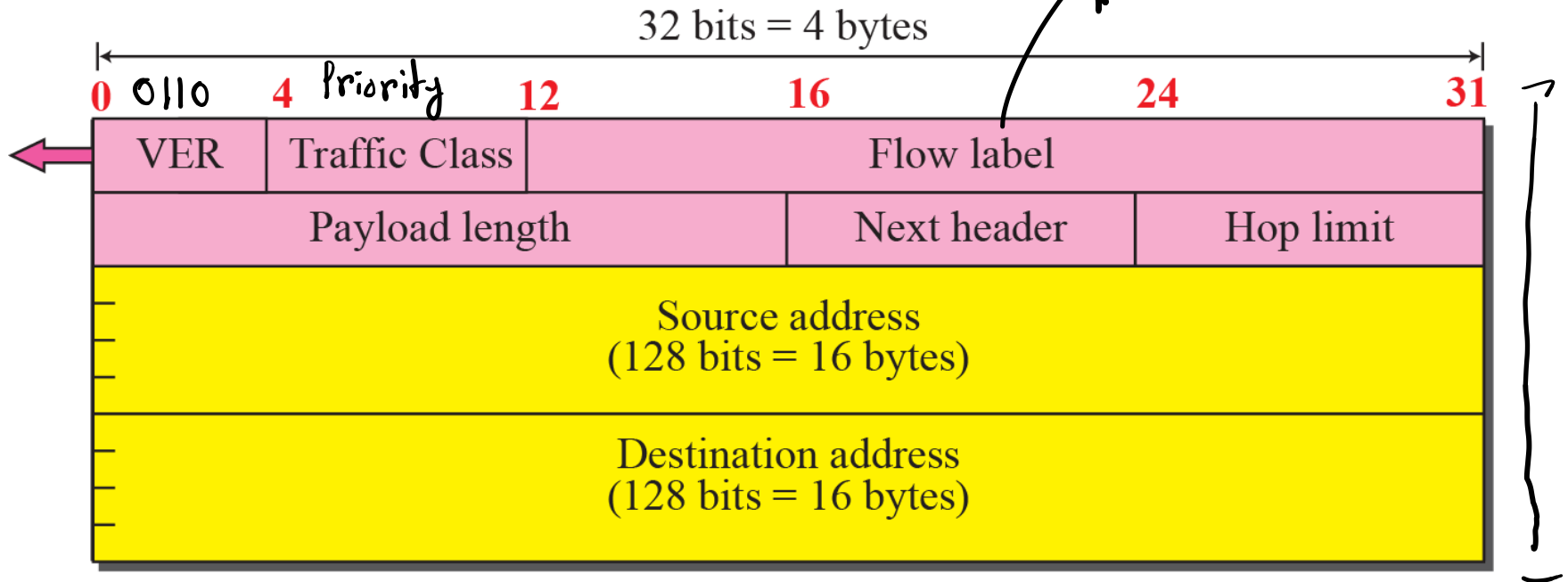
🌱 IPv6 Extension Headers

Optional headers placed after the base header

Examples:

- Hop-by-Hop Options
- Routing
- Fragment
- Authentication
- Encapsulating Security Payload (ESP)

IPv6 Header



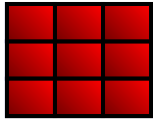
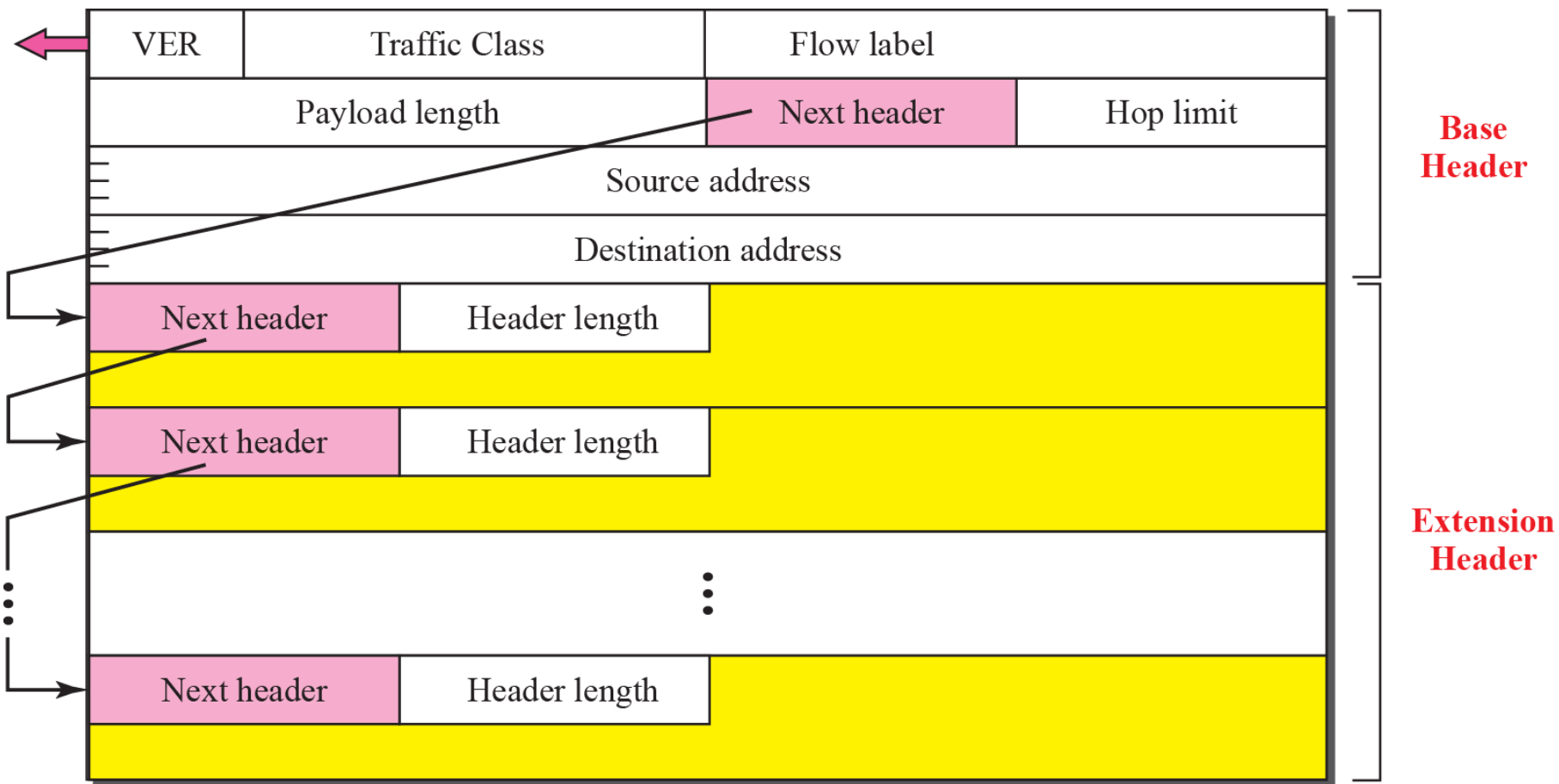
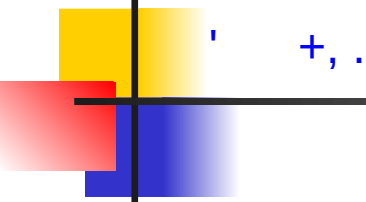
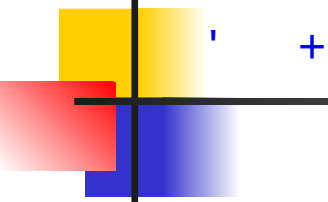


Table 27.1 *Next Header Codes*

<i>Code</i>	<i>Next Header</i>	<i>Code</i>	<i>Next Header</i>
0	Hop-by-hop option	44	Fragmentation
2	ICMP	50	Encrypted security payload
6	TCP	51	Authentication
17	UDP	59	Null (No next header)
43	Source routing	60	Destination option





Extension
headers

Hop-by-hop

Destination

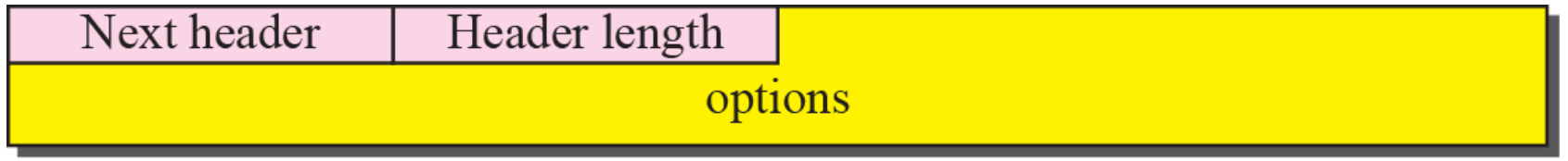
Source routing

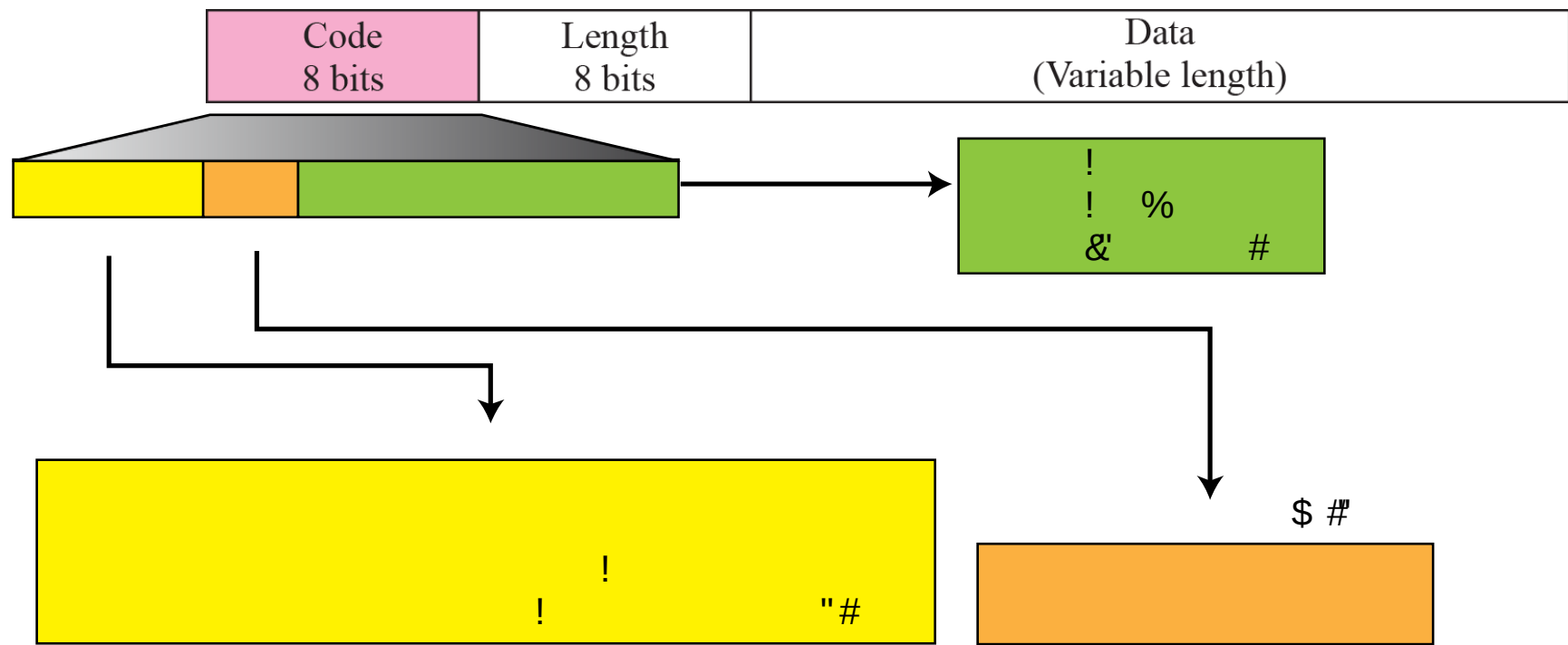
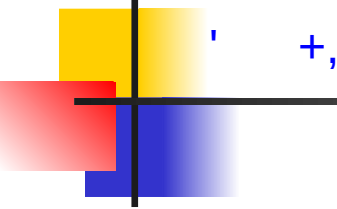
Fragmentation

Authentication

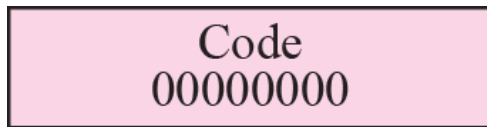
ESP

' +, /

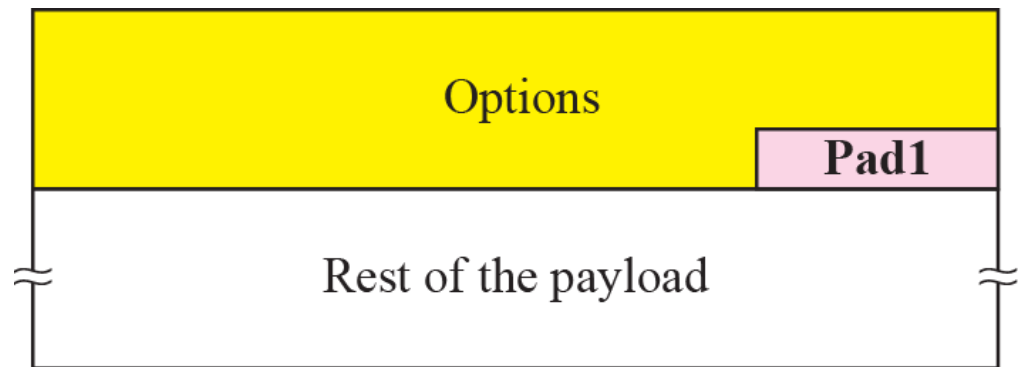




' + , ,



a. Pad1



b. Used for padding

' +, 0

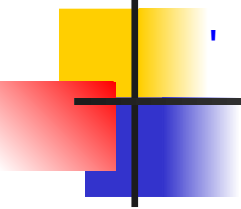


' +, 1

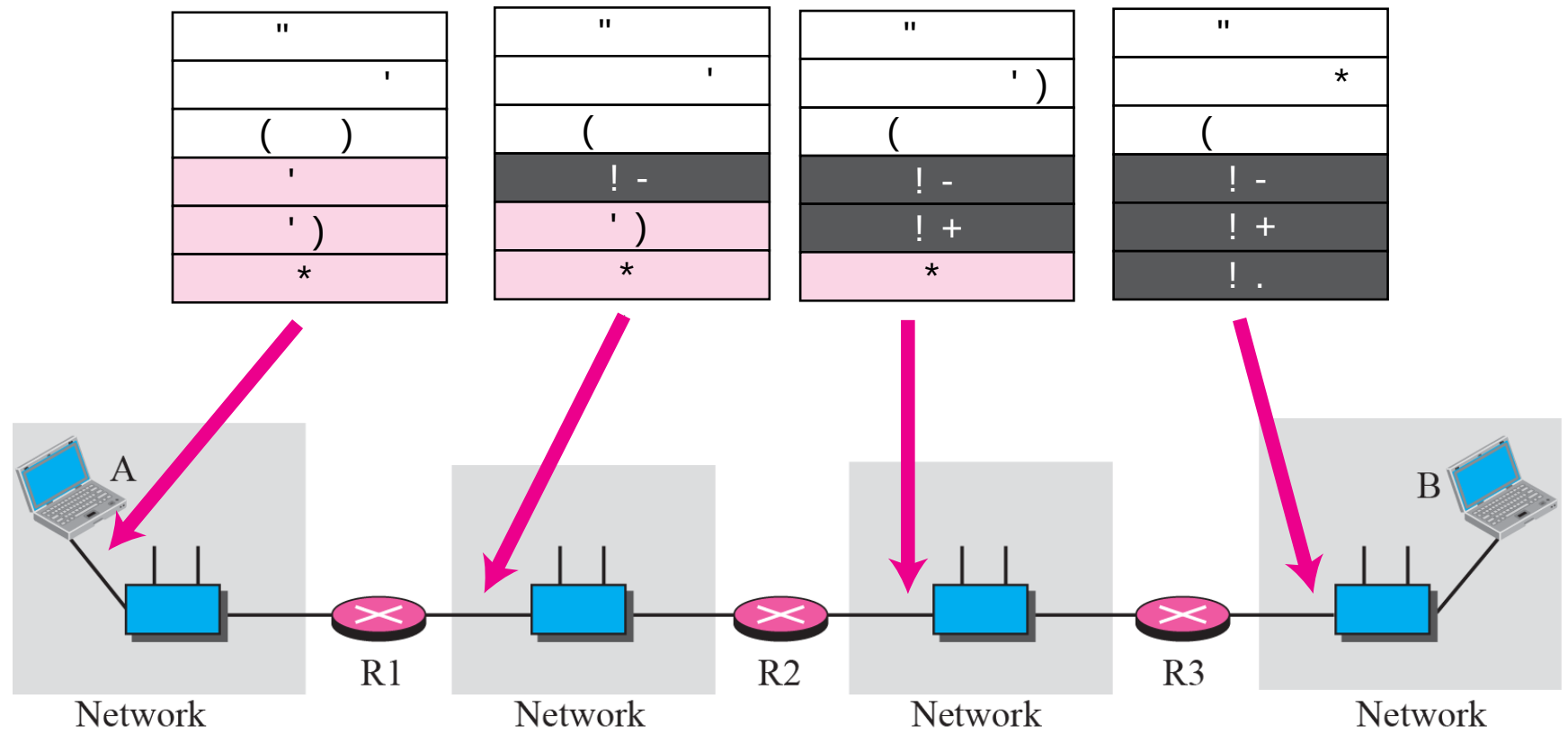
Code		Length
11000010		00000100
Length of jumbo payload 4 bytes		

' +, -2 ! "

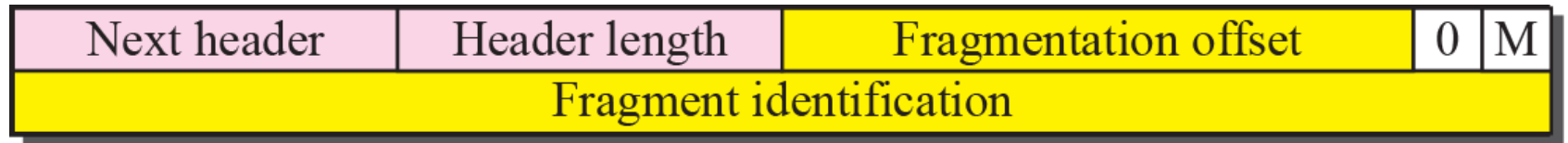
Next header	Header length	Type	Addresses left
Reserved	Strict/loose mask		
First address			
Second address			
⋮			
Last address			



' +, -- ! "



' +, -+

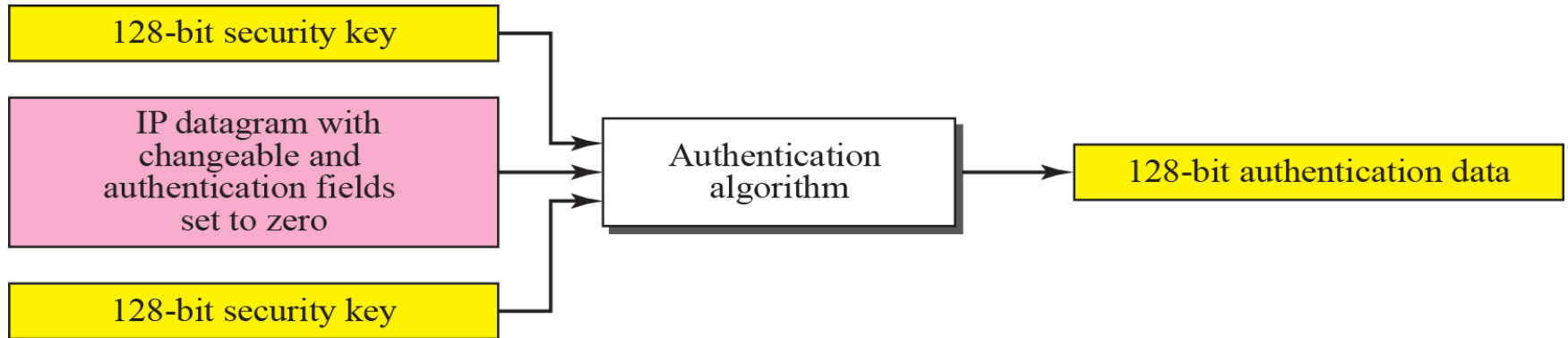


' +, -. # "

Security parameter index

Authentication data

' +, - \$ " "



' +, -/ " "

Security parameter index

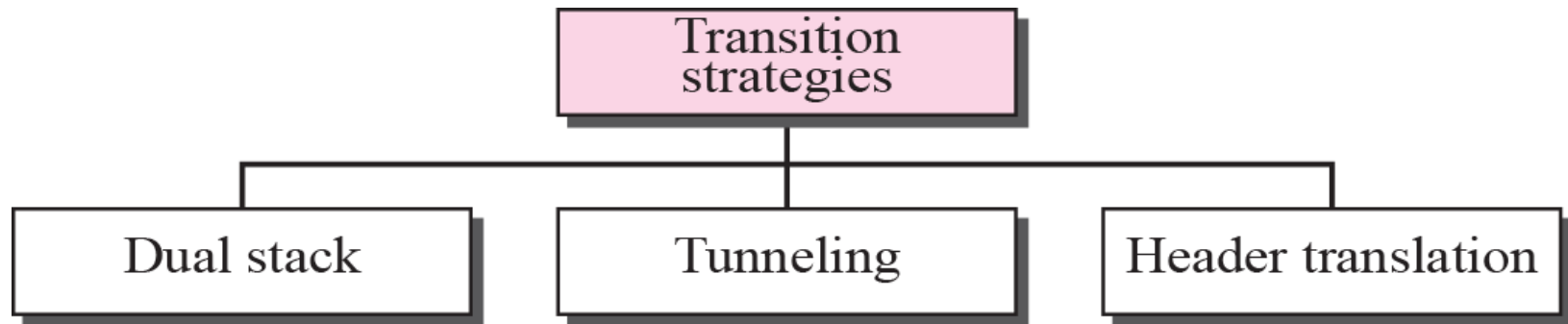
Encrypted data

!

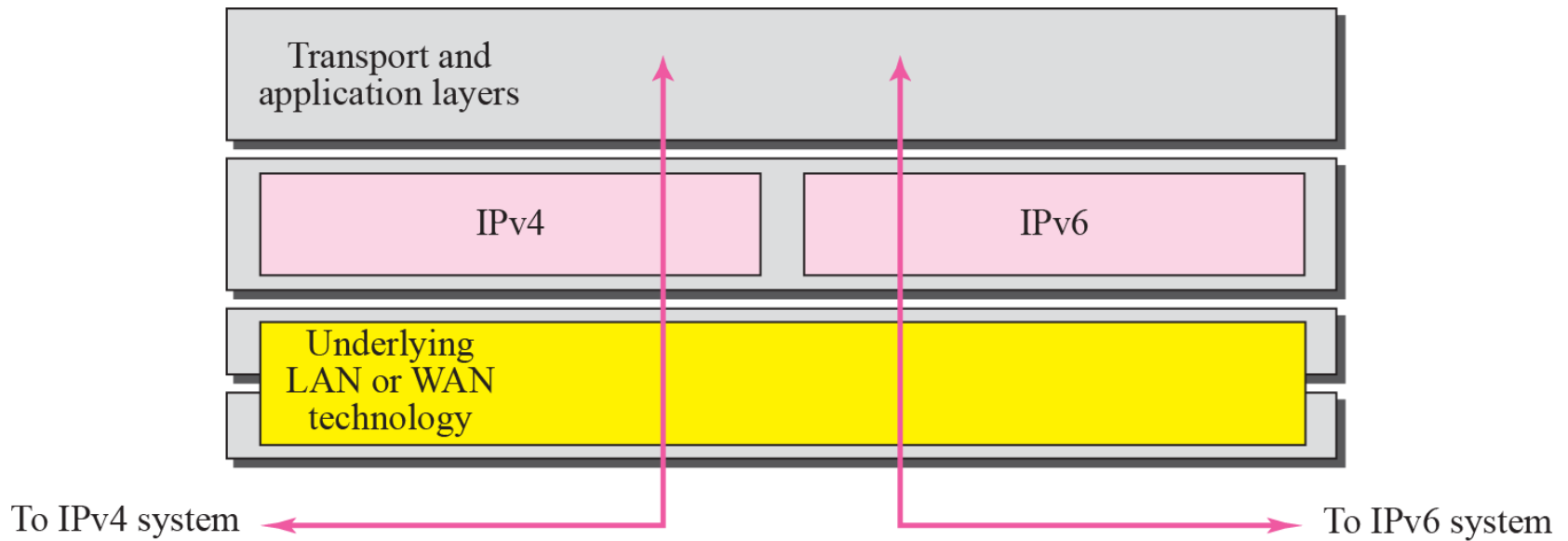
Because of the huge number of systems on the Internet, the transition from IPv4 to IPv6 cannot happen suddenly. It will take a considerable amount of time before every system in the Internet can move from IPv4 to IPv6. The transition must be smooth to prevent any problems between IPv4 and IPv6 systems. Three strategies have been devised by the IETF to help the transition (see Figure 27.16).

3

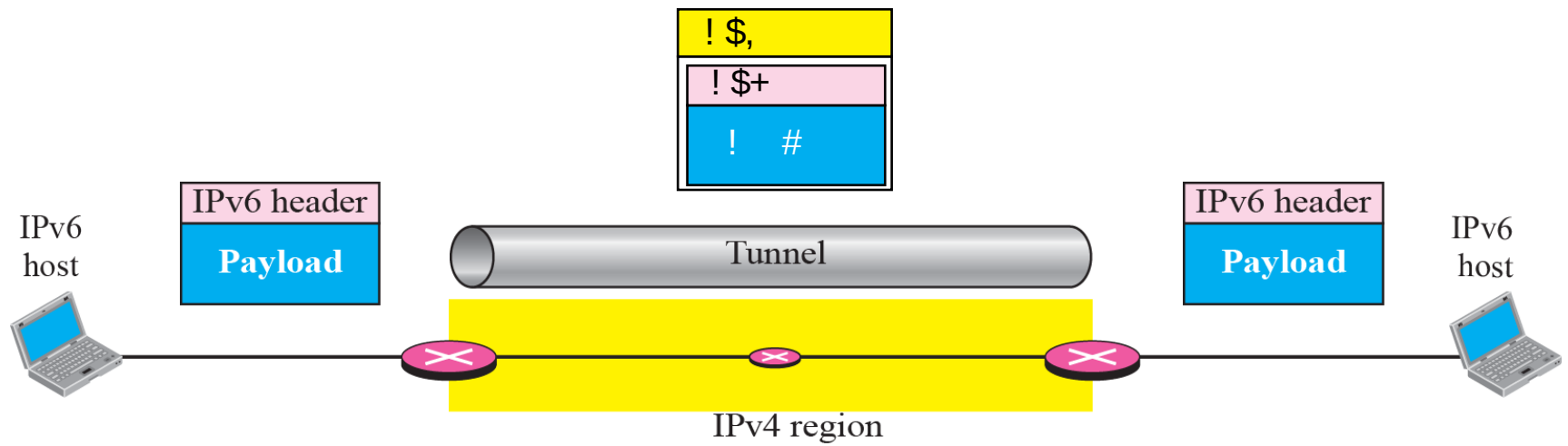
&



' +, -, % "&



+, -0



+, -1

